

12-WEEK PROJECT LOGBOOK

QCampus Connect: Verified Academic Q&A & Secure Communication Platform

Project Title:	QCampus Connect Platform Development
Institution:	University of Energy and Natural Resources (UENR)
Focus Areas:	Web Crypto API, Hyperledger Besu Audit Ledger, BB84 QKD Simulator
Duration:	12 Weeks

WEEK 1

Activities Completed This Week

- Conducted initial project kickoff and stakeholder alignment meetings with project advisors.
- Performed comprehensive requirements gathering and problem formulation surrounding academic trust and accountability in university communication.
- Reviewed existing informal communication platforms and institutional messaging pain points at UENR.
- Drafted initial system requirements specification (SRS) and high-level project roadmap.
- Set up version control repositories (GitHub) and established project collaboration guidelines.

Planned Activity (To Be Done Next Week)

- Finalize detailed software architecture design for QCampus Connect.
- Conduct literature review on Web Crypto API for client-side encryption and Hyperledger Besu for immutable audit logs.
- Define data flow diagrams (DFD) and core entity-relationship (ER) models for users, posts, and verification badges.
- Set up local development environments and tech stack dependencies (Node.js, React, Express, PostgreSQL/MongoDB).

Supervisor's Comments:

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WEEK 2

Activities Completed This Week

- Finalized detailed software architecture design for QCampus Connect.
- Completed theoretical literature review on client-side encryption (Web Crypto API) and private enterprise blockchains (Hyperledger Besu).
- Designed core database schema including verified lecturer profiles, student identity mappings, and specialization badges.
- Established initial development environment, initialized project repositories, and configured boilerplate code.

Planned Activity (To Be Done Next Week)

- Design wireframes and interactive UI/UX mockups for QCampus Connect web interface.
- Establish cryptographic security protocol specifications for SHA-256 message payload hashing.
- Begin development of core authentication module supporting academic email identity verification.

Supervisor's Comments:

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WEEK 3

Activities Completed This Week

- Created UI/UX wireframes and user flow diagrams for verified faculty and student dashboards.
- Defined cryptographic protocol specifications using SHA-256 for data payload verification.
- Implemented base authentication and identity verification services for UENR institutional domain emails.
- Selected and configured frontend component library and state management architecture.

Planned Activity (To Be Done Next Week)

- Develop student and lecturer profile management interfaces with faculty specialization badge display.
- Implement core Q&A forum feed with post creation, categorization, and upvoting features.
- Design client-side encryption module using Web Crypto API for end-to-end direct messaging.

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WEEK 4

Activities Completed This Week

- Developed profile management screens displaying verified academic credentials and faculty specialization badges.
- Built public/academic forum module enabling query posting, topic tagging, and thread navigation.
- Integrated Web Crypto API for generating key pairs on the client side to secure direct messages.
- Engineered backend API endpoints for handling public threads, verified responses, and profile metadata.

Planned Activity (To Be Done Next Week)

- Deploy and configure local Hyperledger Besu private blockchain network node.
- Draft smart contracts for immutable audit logging of academic thread hashes and verification records.
- Connect client-side messaging interface with encrypted direct message routing service.

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WEEK 5

Activities Completed This Week

- Successfully deployed local Hyperledger Besu node for development and testing.
- Developed Solidity smart contracts responsible for recording SHA-256 hashes of academic transactions on-chain.
- Integrated Web Crypto API key exchanges into client-side end-to-end encrypted direct messaging feature.
- Performed initial end-to-end test of encrypted communication between verified user accounts.

Planned Activity (To Be Done Next Week)

- Integrate Hyperledger Besu audit trail logging with the core backend services.
- Build verification component to allow users and auditors to check message/post integrity against blockchain hashes.
- Begin mathematical design and software layout for the educational BB84 Quantum Key Distribution (QKD) simulator module.

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WEEK 6

Activities Completed This Week

- Connected Hyperledger Besu node to main application using web3/ethers middleware.
- Implemented audit trail panel displaying immutable SHA-256 transaction records and timestamps for academic posts.
- Formulated core algorithms for the educational BB84 QKD simulator (quantum state preparation, basis reconciliation, key sift).
- Refined UI components for real-time verification indicators on academic responses.

Planned Activity (To Be Done Next Week)

- Develop full interactive BB84 Quantum Key Distribution simulator UI module within QCampus Connect.
- Implement photon polarization visualization, basis matching table, and eavesdropping (Eve) intercept detection simulation.
- Optimize response latency for Hyperledger Besu transaction logging.

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WEEK 7

Activities Completed This Week

- Built interactive BB84 QKD simulator interface allowing students to model quantum key exchange steps.
- Added simulation controls for basis selection (rectilinear/diagonal), error estimation, and eavesdropper presence.
- Optimized Hyperledger Besu event listeners and transaction submission pipeline for high concurrency.
- Enhanced encrypted direct messaging UI with visual cryptographic status indicators.

Planned Activity (To Be Done Next Week)

- Perform comprehensive integration testing across profile management, Q&A, Web Crypto API, Hyperledger Besu, and BB84 simulator.
- Execute security evaluation on client-side key storage and SHA-256 hash generation mechanisms.
- Identify UI/UX bottlenecks and refine responsive layouts across desktop and tablet views.

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WEEK 8

Activities Completed This Week

- Executed end-to-end integration testing for all system modules and core communication pipelines.
- Conducted security audit on Web Crypto API key handling and encrypted message storage safety.
- Resolved minor UI rendering bugs on the BB84 simulator basis reconciliation visual tables.
- Optimized database queries and API response times for forum search and filtering.

Planned Activity (To Be Done Next Week)

- Conduct functional testing and validation with sample user groups (lecturers and students).
- Verify error handling and fallbacks when Hyperledger Besu node is temporarily unreachable.
- Begin drafting formal technical documentation and user manual for QCampus Connect.

Supervisor's Comments:

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WEEK 9

Activities Completed This Week

- Conducted pilot functional validation testing using simulated academic interaction scenarios.
- Verified system resilience against invalid key entries and missing transaction logs in Hyperledger Besu.
- Gathered preliminary user experience feedback on specialization badge clarity and thread navigation.
- Initiated drafting of system architecture, installation, and deployment chapters of project report.

Planned Activity (To Be Done Next Week)

- Refine system user interface based on pilot feedback, improving accessibility and navigation clarity.
- Detail security analysis chapter including SHA-256 hash integrity and BB84 educational takeaways.
- Prepare system demonstration environment for final evaluation sessions.

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WEEK 10

Activities Completed This Week

- Implemented UI refinements based on pilot feedback, enhancing messaging visual cues and profile layouts.
- Completed security analysis section documenting Web Crypto API mechanisms, Hyperledger audit trail, and BB84 simulation concepts.
- Set up stable local deployment environment and configured seed data for demonstration.
- Conducted secondary regression testing across all core platform features.

Planned Activity (To Be Done Next Week)

- Draft project presentation slides detailing problem statement, architecture, cryptography, and results.
- Finalize complete project documentation draft for preliminary supervisor review.
- Perform final code refactoring and clean up repository codebases.

Supervisor's Comments:

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WEEK 11

Activities Completed This Week

- Created comprehensive presentation slide deck highlighting QCampus Connect architecture, cryptographic framework, and Hyperledger Besu implementation.
- Completed complete draft of project report documentation including system requirements, design diagrams, and evaluation.
- Refactored backend code and finalized inline technical documentation across all modules.
- Submitted project documentation draft to academic supervisor for review and feedback.

Planned Activity (To Be Done Next Week)

- Incorporate supervisor feedback into final project report and slide deck.
- Rehearse project presentation and live demonstration of QCampus Connect and BB84 simulator.
- Prepare final logbook and submission package for academic assessment.

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WEEK 12

Activities Completed This Week

- Incorporated supervisor review edits and finalized complete QCampus Connect project documentation.
- Conduct final dry-run presentation and live system demonstration highlighting blockchain audit log and cryptographic features.
- Finalized weekly logbook entries and compiled complete project deliverable package.
- Successfully presented project defense and submitted final documentation and repository links.

Planned Activity (To Be Done Next Week)

- Project completed. Conduct post-project archive of codebase, database schemas, and presentation materials.
- Document future scalability pathways, including Single Sign-On (SSO) integration and native mobile applications.

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